MDPI

Article

Reconstruction Fidelity and Confidence Reliability in Overlapped Radiation-Detector Pulse Analysis under Distribution Shift

Tao Hong [1,2], Bo Zhang [1,2], Junjie Zhu[3], Yikai Wang[4], Xinjie Yao [5,*] Zhoupeng Guo[6,*] and Guangsheng Xie [7,*]

1 China Nuclear Power Engineering Co., Ltd. Beijing 100875, China
2 CNNC Engineering Research Center for Fuel Reprocessing. Beijing 100875, China
3 School of Artificial Intelligence, Xiangyang Polytechnic University, Xiangyang 441050, China
4 Faculty of Physics and Astronomy, Yunnan University, Kunming 650000, China
5 Faculty of Information Engineering and Automation, Kunming University of Science and Technology. Kunming 650500, China; yaoxinjie@kust.edu.cn
6 School of Automation, Southeast UniversityNanjing 210096China
7 Faculty of Surveying and Information Engineering, West Yunnan University of Applied Sciences, Dali 671000, China
* Correspondence: yaoxinjie@kust.edu.cn; 230259114@seu.edu.cn; guangshengx@wyuas.edu.cn

Abstract

Overlapped radiation-detector pulses must be resolved without conflating accurate pulse counting with reliable recovery of their latent components. This study examines how reconstruction fidelity and predictive confidence respond to controlled changes in detector signals, using the public SAEFit dataset and model as a test case. Component reconstruction was assessed using the labeled semi-synthetic protocol, whereas confidence reliability was evaluated under controlled noise and baseline perturbations. Continued training increased pulse-count accuracy to 99.81%, reduced timing RMSE from 2.177 to 1.308 sampling periods, and raised double-pulse ASNR from 20.621 to 21.174 dB. These gains did not extend to amplitude recovery: amplitude RMSE remained approximately 0.386, compared with the published reference value of 0.056. Under shifted conditions, accuracy decreased to 58.22%, accompanied by pronounced overconfidence. Temperature scaling reduced the expected calibration error from 0.4011 to 0.3273 and the negative log-likelihood from 8.5731 to 2.8681 without altering class decisions. Retaining the most confident 50% of traces increased selective accuracy to 71.91%. These results establish that pulse-count accuracy, component identifiability, and confidence reliability are distinct properties. Calibration improves the interpretation of predictions under distribution shift, but cannot compensate for poorly identified pulse amplitudes.

Keywords: radiation detection; overlapped pulses; pulse decomposition; confidence calibration; distribution shift; selective prediction

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Digital radiation-detector systems record waveform traces from which event timing and deposited energy are inferred for nuclear-decay spectroscopy and related nuclear-structure measurements [1–3]. As detector arrays and acquisition systems operate at increasing count rates, temporally adjacent events may produce partially overlapped pulses [4–6]. Such overlap complicates the extraction of individual pulse parameters and may affect subsequent event identification and spectroscopic interpretation [7]. Reliable

and computationally efficient analysis of overlapped waveforms is therefore an important problem in radiation-sensor signal processing.

Recovering the arrival times and amplitudes of constituent pulses from their superposition is an inverse signal-estimation problem. Its difficulty increases as the temporal separation between pulses decreases and as waveform noise or baseline variation becomes more pronounced. Conventional pulse-shape analysis generally combines a parametric detector-response model with iterative optimization or least-squares estimation [8,9]. These approaches retain a clear relationship between the assumed response function and the estimated physical parameters, but repeated numerical optimization can become computationally demanding for large waveform datasets. Data-driven estimators provide an alternative by learning the mapping from measured waveforms to pulse multiplicity and component parameters, thereby shifting much of the computational cost from inference to model training [10,11].

SAEFit is a recent example of this approach. It jointly performs pulse-count classification, waveform reconstruction, and response-parameter estimation, and uses measured single-pulse waveforms to construct labeled semi-synthetic double-pulse examples [9]. Its public code and data provide a useful basis for examining the behavior of a learned pulse analyzer under a common evaluation protocol. Nevertheless, performance under nominal test conditions does not by itself establish that the resulting predictions remain reliable when the statistical characteristics of the input waveforms change. Variations in noise level and baseline structure can alter both classification performance and the relationship between model confidence and empirical correctness. This issue is particularly relevant when highly confident predictions are used without additional inspection.

Three aspects of performance must therefore be distinguished. The first is pulse-count classification, which determines whether a waveform contains a single or an overlapped event. The second is component-level reconstruction, which concerns the accuracy of the recovered timing and amplitude parameters. The third is confidence reliability, which describes whether the reported class probabilities remain consistent with observed prediction frequencies. These quantities are related but not interchangeable. High pulse-count accuracy does not imply accurate recovery of constituent amplitudes, and improved probability calibration does not correct errors in the reconstructed pulse parameters. Evaluating them separately is necessary to avoid interpreting a strong classification result as evidence of complete waveform decomposition or treating calibrated confidence as evidence of physical reconstruction accuracy.

This distinction is also required by the available ground truth. The public SAEFit release contains measured single-pulse and double-pulse traces, but the measured double-pulse data do not provide independently verified timing and amplitude labels for each constituent pulse. Supervised component-level errors therefore cannot be evaluated on these measured double pulses without introducing additional assumptions or pseudo-labels. Following the original SAEFit protocol, the present study evaluates timing and amplitude recovery using semi-synthetic double pulses generated by shifting, scaling, and summing measured single-pulse waveforms. This construction preserves measured pulse shapes while providing known component parameters, but it does not constitute fully labeled measurement of naturally overlapped events. Accordingly, conclusions concerning component reconstruction are restricted to this semi-synthetic protocol.

Within this evidence boundary, the present study investigates two complementary questions. First, we examine whether continued training and configuration-level adjustment of SAEFit can improve pulse classification and reconstruction while retaining the original data construction, model architecture, and evaluation definitions. Second, we assess the reliability of classification confidence under controlled noise and baseline perturbations.

The perturbations are used as reproducible stress tests rather than as an exhaustive representation of detector-condition changes. Confidence reliability is evaluated using the expected calibration error (ECE), negative log-likelihood (NLL), the frequency of high-confidence errors, and risk–coverage behavior. Post-hoc temperature scaling and confidence-based selective prediction are considered because they modify the interpretation or use of class probabilities without altering the reconstructed waveforms or predicted classes. Monte Carlo dropout is additionally examined as a comparative uncertainty-estimation strategy [12].

The study makes three main contributions. First, it provides an evaluation that explicitly separates nominal component reconstruction from confidence assessment under controlled signal perturbations, thereby preventing conclusions derived from one metric family from being extended to another. Second, the continuation experiments show that pulse-count accuracy, timing error, and reconstructed-waveform quality can improve without a corresponding reduction in component-amplitude error. This result indicates that the remaining amplitude error cannot be attributed solely to insufficient training or temporal misalignment and requires separate methodological attention. Third, the calibration experiments show that post-hoc temperature scaling reduces confidence miscalibration and supports a more informative risk–coverage analysis, while leaving classification accuracy and component estimates unchanged. Taken together, the results demonstrate that reconstruction fidelity and confidence reliability represent distinct properties of an overlapped-pulse analyzer. The conclusions are deliberately confined to the released SAEFit data, the labeled semi-synthetic reconstruction protocol, and the controlled perturbations examined in this study.

2. Related Work

2.1. Pulse analysis and detector signal processing

Digital detector systems have substantially increased the throughput and temporal resolution of nuclear spectroscopy experiments[4,5]. Further optimization of digital acquisition systems for fast scintillator detectors has increased both the volume of recorded waveforms and the demand for efficient waveform processing[6]. Under high-count-rate operation, pulse pile-up and waveform overlap become important sources of uncertainty because a single recorded trace may contain contributions from more than one physical pulse. Pulse-shape analysis is therefore commonly formulated as an inverse estimation problem in which timing, amplitude, and related response parameters are inferred from sampled detector waveforms.

Classical approaches usually rely on response-function modeling and iterative parameter fitting. These methods are attractive because their assumptions are explicit and their estimated parameters can often be interpreted with respect to a physical or instrumental response model. Their limitations are also well known: nonlinear or least-squares fitting can be computationally expensive for large waveform streams, and the resulting estimates may depend on the adequacy of the assumed response model and initialization strategy[9,13]. These limitations have motivated learned waveform analyzers that approximate the mapping from detector traces to pulse classes and pulse parameters more efficiently[8].

SAEFit is a representative recent framework in this direction. It uses a learned pulse-analysis model to classify pulse multiplicity and estimate response parameters for single and double pulses[9]. Its importance for the present study lies not only in the model itself, but also in the availability of a public dataset and reproducible evaluation protocol. This makes SAEFit a useful basis for examining how reconstruction metrics and reliability metrics should be reported for overlapped radiation-detector pulses.

A methodological distinction is needed between measured waveforms and labeled decomposition targets. The public SAEFit release includes measured single-pulse traces and measured real double-pulse traces, but the real double-pulse traces do not provide independently observed component times and amplitudes. Component-level supervised metrics therefore depend on the semi-synthetic protocol in which measured single-pulse waveforms are shifted, scaled, and summed according to explicit rules. This protocol preserves measured pulse shapes while providing known labels, but it should not be interpreted as fully labeled real double-pulse supervision. Accordingly, component RMSE values should be reported as protocol-dependent reconstruction metrics, whereas unlabeled real double-pulse traces can support waveform-level or distributional checks but not supervised component-accuracy claims.

2.2. Neural estimators and optimization

Learned estimators have become common in signal-processing tasks where high-dimensional observations must be mapped to task-specific quantities[11]. Residual architectures and attention-based models have shown that learned representations can reduce the need for hand-crafted features in many domains[14,15]. Probabilistic output layers and mixture-based models further provide mechanisms for representing uncertainty or ambiguity in predicted targets[16,17]. For detector-pulse analysis, however, model capacity alone is not sufficient. Since timing and amplitude estimates may enter downstream physical interpretation, improvements in learned representations must be evaluated against clearly defined measurement targets and label sources.

The present work does not introduce a new pulse model, architecture, or loss function. Instead, it examines whether an existing SAEFit checkpoint can be improved through score-preserving continuation while keeping the public evaluation basis unchanged. The adjustable factors are limited to already supported training controls, including learning rate, loss weights, continuation budget, and checkpoint selection. This design is intentionally conservative: it keeps the comparison close to the public baseline and reduces the risk that an apparent improvement is caused by changes in data splits, labels, or metric definitions.

The optimization procedures used in this setting are standard components of modern neural training. Adam-style optimizers, decoupled weight decay, and widely used PyTorch-based implementations provide the computational infrastructure for continuation experiments[18–21]. We therefore treat optimization as a controlled experimental procedure rather than as the primary methodological contribution. This distinction is important for reproducibility, because the scientific claim depends on whether reconstruction and calibration results are obtained under fixed evaluation semantics rather than on the novelty of the optimizer or implementation stack.

2.3. Uncertainty, calibration, and abstention

Uncertainty estimation and calibration are closely related but distinct aspects of reliable prediction. Predictive uncertainty can be approximated using methods such as Bayesian dropout, deep ensembles, and heteroscedastic regression[12,22–24]. Post-hoc calibration methods provide a complementary approach by adjusting predictive confidence after model training, without necessarily changing the predicted class or regression output[25–27]. These methods are particularly relevant when a model is already trained and the main question is whether its reported confidence is aligned with observed correctness.

The statistical basis of calibration predates modern neural models. Well-calibrated probabilistic prediction has been studied from Bayesian and frequentist perspectives[28], and probability-score decomposition provides a principled way to separate different aspects of forecast quality[29]. Reliability diagrams and related calibration diagnostics remain

useful tools for assessing whether predicted confidence values correspond to empirical accuracy[30]. In sensor applications, this distinction is important because a prediction may be numerically accurate under nominal conditions while becoming overconfident under shifted operating conditions.

Distribution shift is a central challenge for learned sensor systems. Changes in noise level, baseline behavior, environmental conditions, or acquisition electronics can alter the input distribution encountered during deployment. Prior studies have shown that such variability can degrade model robustness and confidence reliability[31,32]. Recent sensor-oriented work has therefore considered uncertainty-aware or distribution-aware decision mechanisms under changing conditions[33]. For radiation-detector pulse analysis, the same concern arises when a model trained or tuned under one waveform condition is applied to traces with different noise or baseline characteristics.

Reliable confidence estimates can also support abstention. Selective classification formalizes the trade-off between coverage and risk by allowing a model to withhold predictions on examples judged insufficiently reliable[34,35]. Conformal prediction provides a related framework for distribution-free uncertainty quantification under appropriate assumptions[36,37], and recent sensor studies have used conformal calibration to construct reliable prediction intervals[38]. In the present study, selective prediction is used more modestly as a risk–coverage diagnostic rather than as a formal conformal guarantee.

Taken together, these studies motivate a cautious evaluation strategy. Calibration and abstention can improve how confidence is reported and used, but they do not repair inaccurate physical decomposition. A detector-pulse analyzer that performs well on clean validation traces may still assign excessive confidence to shifted or ambiguous traces, while a calibrated confidence score does not prove that subpulse amplitudes have been accurately recovered. The present work therefore treats uncertainty calibration as a decision layer on top of a fixed decomposition model and reports it separately from component-level reconstruction performance.

3. Methods

3.1. Study design and evidence boundary

The study was designed to evaluate two distinct properties of the SAEFit pulse analyzer: component-reconstruction fidelity under the original evaluation protocol and classification-confidence reliability under controlled signal perturbations. These properties were assessed separately because the available reference information differs between the two settings and because confidence calibration does not alter the reconstructed pulse parameters.

For a normalized detector waveform $\mathbf{x}$, SAEFit produces class logits $\mathbf{z}$, a reconstructed waveform $\hat{\mathbf{x}}$, and estimated response parameters describing pulse timing, amplitude, and shape [9]. The classifier distinguishes single- from double-pulse traces, whereas the reconstruction and parameter-estimation outputs are used to evaluate waveform and component-level fidelity.

The public data contain measured single-pulse and double-pulse traces. However, the measured double-pulse traces do not include independently verified timing and amplitude labels for each constituent pulse. Component-level errors therefore cannot be calculated for these traces without introducing additional assumptions. Following the original SAEFit protocol, labeled double pulses are constructed by shifting, scaling, and summing measured single-pulse waveforms. The resulting semi-synthetic traces retain measured pulse shapes while providing known component timing and amplitude values. Table 1 summarizes the resulting evidence boundary.

Table 1. Data sources, available reference information, and supported evaluations.

Data category	Waveform origin	Available reference	Supported evaluation
Measured single pulses	Detector measurements	Single-pulse class and measured waveform	Pulse-count classification and single-pulse reconstruction
Semi-synthetic double pulses	Transformed and summed measured single pulses	Component timing and amplitude defined by construction	Double-pulse classification, waveform reconstruction, and component-level errors
Measured double pulses	Detector measurements	Double-pulse waveform without independent component labels	Waveform-level assessment; no supervised component RMSE
Perturbed evaluation traces	Controlled perturbations applied to evaluation waveforms	Retained pulse-count labels	Shifted classification, calibration, and selective-risk assessment

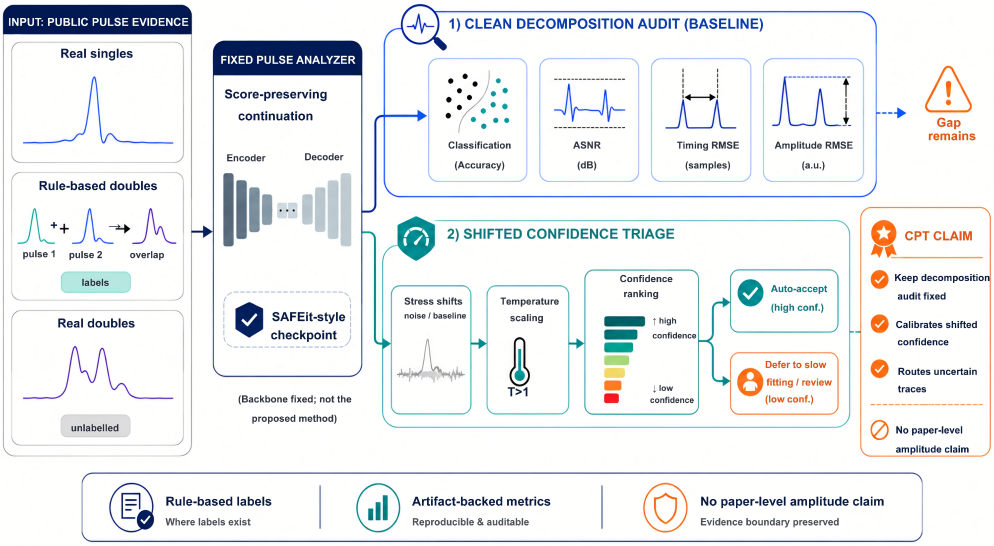


Figure 1. Overview of the evaluation design. Pulse classification and component reconstruction are assessed using the unchanged SAEFit data protocol, whereas confidence reliability is evaluated separately under controlled noise and baseline perturbations. Post-hoc calibration and confidence-based selection operate only on classification probabilities and do not modify the reconstructed waveforms or estimated pulse parameters.

Accordingly, timing and amplitude RMSE are reported only for the labeled semi-synthetic protocol. Controlled noise and baseline perturbations are used only to examine pulse-count classification and confidence behavior. They are not used to infer component-level reconstruction errors. Figure 1 summarizes this evaluation design.

3.2. Protocol-preserving model continuation

The continuation experiments examine whether additional training and adjustment of existing optimization controls improve SAEFit performance without changing the underlying problem definition. The model architecture, waveform-construction procedure, data split, target labels, and metric implementations are retained from the public SAEFit protocol. The varied quantities are restricted to training controls already supported by the implementation: learning rate, continuation duration, checkpoint selection, the timing-loss weight τ, the classification-loss weight τ_c, and the reconstruction-related weight λ.

Continuation is performed in two stages. The first stage prioritizes temporal localization by increasing the contribution of the timing objective while reducing the weight assigned to the already high-performing pulse-count classifier. Timing RMSE is treated as the primary quantity for checkpoint comparison in this stage, while classification accuracy and waveform-reconstruction metrics are monitored for deterioration. The second

stage examines whether adjustment of the reconstruction-related loss weight improves component-amplitude estimation without substantially degrading the other reported metrics. Amplitude RMSE is treated as the primary quantity in this stage.

This procedure is not introduced as a new learning algorithm. It is a controlled examination of the extent to which the performance of an existing SAEFit model can change under additional training and configuration-level adjustment. All comparisons use the same evaluation definitions, and results are reported for the full metric set rather than only for the metric prioritized during a given stage. This prevents an improvement in timing or classification from being interpreted as a general improvement in component reconstruction.

3.3. Controlled signal-perturbation protocol

Confidence reliability is evaluated using perturbed versions of normalized evaluation traces. Let $\mathbf{x}_i$ denote an unperturbed normalized waveform. A perturbed waveform is defined as

$$\mathbf{x}_i^{(s)} = \mathbf{x}_i + \boldsymbol{\eta}_i^{(s)} + \mathbf{b}_i^{(s)}, \tag{1}$$

where $\boldsymbol{\eta}_i^{(s)}$ denotes additive waveform noise, $\mathbf{b}_i^{(s)}$ denotes a baseline perturbation, and s identifies the perturbation condition. The perturbations are applied after the original waveform normalization and before model inference. Five conditions are considered: noise only, baseline only, a reference combined condition, a mild combined condition, and a strong combined condition. The corresponding perturbation magnitudes and sampling procedures are specified in the experimental protocol.

The original pulse-count label is retained because the perturbations are assumed not to change whether the underlying waveform contains one or two pulses. This assumption defines the scope of the stress test. The perturbed traces represent controlled departures from the nominal evaluation distribution; they are not claimed to reproduce the full range of changes that may occur in detector response, electronics, sampling, or experimental conditions.

For each perturbed waveform, the model produces logits $\mathbf{z}_i$. The probability assigned to class k is

$$p_{ik} = \frac{\exp(z_{ik})}{\sum_{j=1}^{K} \exp(z_{ij})}, \tag{2}$$

where $K = 2$ for single- and double-pulse classification. The predicted class and associated confidence are

$$\hat{y}_i = \arg\max_k \, p_{ik}, \qquad c_i = \max_k \, p_{ik}. \tag{3}$$

No perturbed trace is used to calculate component timing or amplitude RMSE.

3.4. Reconstruction and classification metrics

Let $\delta_{\hat{y}_i, y_i}$ denote the Kronecker delta, which equals 1 when $\hat{y}_i = y_i$ and 0 otherwise. Pulse-count accuracy is calculated as

$$\mathrm{Acc} = \frac{1}{n} \sum_{i=1}^{n} \delta_{\hat{y}_i, y_i}, \tag{4}$$

where y_i is the reference pulse-count label.

Following the public SAEFit evaluation convention, waveform-reconstruction quality is summarized using the average signal-to-noise ratio (ASNR). Let s_{ij} and $\hat{s}_{ij}$ denote the target and reconstructed values, respectively, at sample j of trace i. The trace-level signal-to-noise ratio is

$$\mathrm{SNR}_i = 10 \log_{10} \left[\frac{\sum_{j=1}^{L} s_{ij}^2}{\sum_{j=1}^{L} \left(s_{ij} - \hat{s}_{ij} \right)^2} \right], \tag{5}$$

where L is the number of waveform samples. ASNR is the mean over all evaluated traces:

$$\mathrm{ASNR} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{SNR}_i. \tag{6}$$

ASNR is reported in decibels, with larger values indicating closer waveform agreement. Separate values are reported for single pulses, individual subpulses, and reconstructed double pulses when the corresponding reference waveform is available.

For a component-level target q_i and its estimate $\hat{q}_i$, RMSE is defined as

$$\mathrm{RMSE}(q, \hat{q}) = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (q_i - \hat{q}_i)^2}. \tag{7}$$

Timing RMSE is expressed in sampling periods, whereas amplitude RMSE is calculated using normalized component amplitudes. For double pulses, amplitude errors are reported separately for the two ordered components. Because these quantities require component labels, they are evaluated only on semi-synthetic double pulses.

3.5. Confidence-calibration metrics

Confidence reliability under controlled perturbation is assessed using the expected calibration error (ECE), negative log-likelihood (NLL), and overconfident error rate. For ECE, predictions are partitioned into B confidence intervals, denoted by $\mathcal{I}_1, \ldots, \mathcal{I}_B$. The empirical accuracy of bin $\mathcal{I}_b$ is

$$\mathrm{acc}(\mathcal{I}_b) = \frac{1}{|\mathcal{I}_b|} \sum_{i \in \mathcal{I}_b} \delta_{\hat{y}_i, y_i}, \tag{8}$$

and its mean confidence is

$$\mathrm{conf}(\mathcal{I}_b) = \frac{1}{|\mathcal{I}_b|} \sum_{i \in \mathcal{I}_b} c_i. \tag{9}$$

ECE is calculated as

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|\mathcal{I}_b|}{n} |\mathrm{acc}(\mathcal{I}_b) - \mathrm{conf}(\mathcal{I}_b)|. \tag{10}$$

Because ECE depends on the binning scheme, the number and definition of the confidence bins are held fixed for all comparisons.

NLL is defined as

$$\mathrm{NLL} = -\frac{1}{n} \sum_{i=1}^{n} \log p_{i, y_i}. \tag{11}$$

Unlike classification accuracy, NLL depends on the probability assigned to the reference class and therefore penalizes confident errors.

For a confidence threshold ρ, define the overconfident-error indicator as

$$r_i(\rho) = \begin{cases} 1, & \hat{y}_i \neq y_i \text{ and } c_i \geq \rho, \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

The overconfident error rate is then

$$\mathrm{OER}_\rho = \frac{1}{n} \sum_{i=1}^{n} r_i(\rho). \tag{13}$$

The experiments use $\rho = 0.85$. The denominator is the total number of evaluated traces, ensuring that values remain comparable across perturbation conditions.

3.6. Post-hoc temperature scaling

Temperature scaling is applied to the classification logits using a scalar parameter $T > 0$ [27]. The temperature is estimated on the perturbed validation subset by minimizing NLL:

$$T^* = \arg\min_{T>0} \left\{ -\frac{1}{n_{\mathrm{val}}} \sum_{i=1}^{n_{\mathrm{val}}} \log \left[\frac{\exp(z_{i,y_i}/T)}{\sum_{k=1}^{K} \exp(z_{ik}/T)} \right] \right\}. \tag{14}$$

The calibrated probability assigned to class k for a held-out evaluation trace is

$$p_{ik}^{(T)} = \frac{\exp(z_{ik}/T^*)}{\sum_{j=1}^{K} \exp(z_{ij}/T^*)}. \tag{15}$$

For $T^* > 0$, temperature scaling preserves the ordering of the logits and therefore does not change the predicted class. Differences between calibrated and uncalibrated results are consequently restricted to confidence-dependent quantities such as ECE, NLL, the overconfident error rate, and selective risk. Temperature scaling does not modify waveform reconstruction or component-parameter estimates.

As a secondary comparison, Monte Carlo dropout retains dropout during inference and averages the class probabilities obtained from repeated stochastic forward passes [12]. The number of forward passes and retained dropout configuration are reported in the experimental protocol. This comparison is used only to assess classification-confidence behavior.

3.7. Selective prediction

Selective prediction is evaluated by ranking traces according to their maximum calibrated confidence:

$$c_i^{(T)} = \max_k p_{ik}^{(T)}. \tag{16}$$

For target coverage γ, let $\mathcal{A}_\gamma$ contain the $\lceil \gamma n \rceil$ traces with the largest calibrated confidence values. Selective accuracy is

$$\mathrm{Acc}_\gamma = \frac{1}{|\mathcal{A}_\gamma|} \sum_{i \in \mathcal{A}_\gamma} \delta_{\hat{y}_i, y_i}, \tag{17}$$

and selective risk is

$$\mathrm{Risk}_\gamma = 1 - \mathrm{Acc}_\gamma. \tag{18}$$

The resulting risk–coverage curve assesses whether confidence ranking identifies subsets with different empirical error rates.

This procedure is a selective-classification diagnostic and is not a conformal-prediction method. It does not construct prediction sets, establish finite-sample coverage, or provide a distribution-free guarantee. The reported coverage denotes the proportion of traces retained after confidence ranking rather than a guaranteed probability of containing a correct label.

3.8. Protocol control and reproducibility

All nominal reconstruction comparisons retain the public SAEFit data split, semi-synthetic waveform-construction rules, label definitions, and metric implementations. Training configurations and selected checkpoints are recorded for each continuation experiment. Calibration parameters are estimated using validation predictions and evaluated using held-out predictions. The uncalibrated and calibrated metrics are computed from the same logits so that temperature scaling cannot change the underlying classification decisions.

Results that do not support the intended hypothesis are retained in the analysis. In particular, component-amplitude RMSE is reported alongside timing and waveform metrics, and Monte Carlo dropout is not treated as the principal calibration result when its observed improvement is limited. The absence of component labels for measured double pulses is handled by restricting the corresponding conclusions rather than introducing pseudo-labels. These controls ensure that reported improvements are attributable to the specified training or calibration operation and do not result from changes in the evaluation definition.

4. Experimental Protocol

4.1. Data use and reference models

All experiments were conducted using the public SAEFit code and data release [9]. The original waveform preprocessing, data partition, semi-synthetic pulse-construction procedure, and metric implementations were retained. No additional detector dataset or independently generated component label was introduced.

The public data were used according to the availability of reference labels. Measured single-pulse traces were used directly and were also shifted, amplitude-scaled, and summed according to the public SAEFit procedure to construct labeled semi-synthetic double pulses. These semi-synthetic traces provide known component delays and amplitudes and therefore support supervised timing and amplitude evaluation. The measured double-pulse traces do not contain independently verified labels for the constituent timing and amplitude parameters. They were consequently excluded from supervised component-level RMSE calculations.

Three model-reference levels were distinguished. The first was the locally reproduced epoch-125 SAEFit checkpoint, which provided the initial reference for the continuation experiments. It achieved a pulse-count accuracy of 99.40%, a single-pulse ASNR of 21.12 dB, subpulse ASNR values of 16.76 and 16.89 dB, and a double-pulse ASNR of 20.06 dB. The second reference consisted of the values reported for the 750-epoch model in the original SAEFit study: 99.63% pulse-count accuracy, 21.57 dB single-pulse ASNR, 20.20 and 20.32 dB subpulse ASNR, 21.94 dB double-pulse ASNR, a timing RMSE of 0.71 sampling periods, and amplitude RMSE values of 0.058 and 0.056. These published values were used only

Table 2. Selected configurations for the two continuation stages.

Stage	Initial checkpoint	τ	τ_c	λ	Learning rate	Primary evaluation quantity
Timing-focused	Continued checkpoint near epoch 500	5.0	0.05	0.01	3×10^{-4}	Timing RMSE
Amplitude-focused	Selected timing-focused checkpoint	3.0	0.05	0.05	2×10^{-4}	Amplitude RMSE

for contextual comparison and were not re-estimated or used as training targets. The third reference was the final checkpoint selected from the continuation experiments described below.

4.2. Continuation and configuration selection

The epoch-125 checkpoint was first continued toward approximately 500 training epochs while retaining the original model architecture and evaluation procedure. A checkpoint near the end of this continuation stage was then used to initialize two sequential configuration stages. The first stage prioritized timing estimation, whereas the second examined whether reconstruction-related weighting could improve component-amplitude estimation.

The selected configurations are summarized in Table 2. Apart from the listed parameters, the optimizer, scheduler, data construction, and remaining training settings were retained from the public implementation.

The timing-focused stage increased the contribution of the timing objective while reducing the weight assigned to the already high-performing classifier. Checkpoint comparison primarily considered timing RMSE, while classification accuracy and ASNR were monitored to identify possible deterioration in other outputs. The amplitude-focused stage started from the selected timing-focused checkpoint and increased the reconstruction-related weight. Amplitude RMSE was treated as the primary evaluation quantity, with timing, classification, and ASNR retained as complementary metrics.

The configuration search was deliberately bounded and was not intended to constitute an exhaustive hyperparameter optimization. The final amplitude-focused run was evaluated after 40 continuation epochs. Only completed runs are reported; no result was extrapolated from an unexecuted or unfinished configuration. Because model and checkpoint selection used feedback from the public evaluation protocol, the selected values are interpreted as optimized results within that protocol rather than as independent estimates of performance on a new detector population.

4.3. Controlled signal-perturbation protocol

Confidence reliability was assessed using controlled perturbations applied to normalized evaluation traces. The perturbations were introduced after waveform normalization and before model inference. The pulse-count labels were retained because the perturbations were designed to modify signal quality without changing whether the underlying trace contained one or two pulses.

Five perturbation conditions were considered:

1. additive waveform noise only;
2. baseline variation only;
3. a reference combined noise-and-baseline condition;
4. a mild combined condition; and
5. a strong combined condition.

The same final continued checkpoint was evaluated under every condition. Perturbed traces were not used to retrain the decomposition model and were not used to calculate component timing or amplitude RMSE. The perturbation experiments were restricted to pulse-count accuracy and confidence-dependent metrics, including ECE, NLL, the overconfident error rate at a confidence threshold of 0.85, and selective risk.

These perturbations should be interpreted as controlled signal-level stress tests. They do not represent an exhaustive model of detector drift and do not include all possible changes in detector response, electronics, pulse shape, pile-up distribution, or sampling behavior. Conclusions concerning distribution shift are therefore restricted to the perturbation types and intensities evaluated in this study.

4.4. Confidence calibration and selective prediction

For each perturbation condition, uncalibrated class probabilities were obtained from the softmax outputs of the selected checkpoint. Temperature scaling was fitted using the corresponding validation logits by minimizing NLL. The selected temperature was $T = 8.0$. This scalar was subsequently applied to the held-out evaluation logits.

Because division by a positive temperature preserves the ordering of class logits, temperature scaling does not change the predicted pulse-count class. Calibrated and uncalibrated accuracy must therefore be identical when they are calculated from the same logits. The comparison was restricted to confidence-sensitive quantities, namely ECE, NLL, the overconfident error rate, and risk–coverage behavior.

Monte Carlo dropout was examined as a secondary uncertainty-estimation comparison. Dropout was retained during inference, and predictive probabilities were obtained by averaging repeated stochastic forward passes. This operation affected only the classification probabilities and did not alter the waveform-reconstruction or component-parameter outputs. Because the observed changes were limited, Monte Carlo dropout was retained as a comparative result rather than used as the principal calibration method.

Selective prediction was evaluated by ranking traces according to maximum temperature-scaled confidence. For each target coverage, the highest-confidence fraction of traces was retained, and selective accuracy and selective risk were calculated for that subset. Coverage therefore denotes the empirical proportion of retained traces. This experiment provides a risk–coverage diagnostic and does not constitute conformal prediction or provide a distribution-free coverage guarantee.

4.5. Comparison strategy

The three reference levels address different experimental questions and were therefore not combined into a single aggregate score. Comparison with the epoch-125 checkpoint assesses whether continued training and configuration adjustment improve the local SAE-Fit reproduction. Comparison with the published 750-epoch values indicates how closely the continued model approaches the performance reported in the original study. The calibration comparison assesses whether post-hoc scaling improves the statistical reliability of classification confidence under the tested perturbations without altering the decomposition checkpoint.

The local epoch-125 checkpoint and the published 750-epoch reference originate from different training histories. Their comparison is therefore descriptive rather than a paired statistical comparison. Similarly, exceeding a published value for one metric does not establish general superiority when other reconstruction quantities remain below the published reference. Classification accuracy, ASNR, timing RMSE, amplitude RMSE, and calibration metrics are consequently reported separately.

Table 3. Performance under the nominal SAEFit evaluation protocol. ASNR values are reported in decibels, timing RMSE in sampling periods, and amplitude RMSE in normalized amplitude units. A dash indicates that the corresponding value was not available.

Checkpoint	Acc. (%)	$ASNR_s$	$ASNR_{p1}$	$ASNR_{p2}$	$ASNR_d$	$RMSE_t$	$RMSE_{a,2}$
Epoch 125	99.40	21.12	16.76	16.89	20.06	–	–
Published 750-epoch reference	99.63	21.57	20.20	20.32	21.94	0.710	0.0560
Continued checkpoint	99.77	21.16	18.59	18.16	20.62	2.177	0.3869
Timing-focused checkpoint	99.80	21.29	19.17	18.79	21.10	1.402	0.3866
Amplitude-focused checkpoint	99.81	–	19.27	18.80	21.17	1.308	0.3859

The uncertainty analysis is also interpreted independently of the nominal reconstruction comparison. A reduction in ECE or NLL does not imply improved component timing or amplitude estimation, and an unchanged shifted accuracy does not invalidate a calibration improvement. The two analyses address different properties of the same checkpoint.

4.6. Implementation and reproducibility controls

The continuation experiments used the public SAEFit training stack and retained the original model interface, optimizer family, scheduler, waveform-construction procedure, and metric implementations. No additional neural module was introduced. Configuration changes were restricted to the parameters reported in Table 2.

For calibration, the trained checkpoint was loaded once and the classification logits were recorded for the validation and evaluation partitions. The same evaluation logits were used for calibrated and uncalibrated comparisons. This paired construction ensures that any difference in confidence-sensitive metrics results from probability rescaling rather than from different input traces or classification decisions.

Quantitative figures were generated programmatically from the recorded numerical results. For the target-normalized reconstruction plot presented in the Results section, higher-is-better quantities were normalized as the observed value divided by the published reference value. Lower-is-better quantities were normalized as the published reference value divided by the observed value. A normalized value of one therefore indicates equality with the published reference for either type of metric. Exact unnormalized values are reported in the corresponding tables and should be used for quantitative interpretation.

All reported reconstruction metrics were calculated using unchanged target definitions. Unfavorable results were retained when they constrained the interpretation of the model. In particular, amplitude RMSE was reported alongside timing and ASNR, and the absence of component labels for measured double pulses was handled by restricting the evaluation scope rather than by introducing pseudo-labels. These controls prevent calibration improvements from being interpreted as evidence of improved pulse decomposition.

5. Results

5.1. Reconstruction performance under the nominal protocol

Table 3 reports pulse-count classification, waveform reconstruction, timing estimation, and second-component amplitude estimation under the nominal SAEFit evaluation protocol. The published 750-epoch values are included as contextual references rather than as results reproduced in the present study.

Relative to the continued checkpoint, timing-focused continuation reduced timing RMSE from 2.177 to 1.402 sampling periods. It also increased the first- and second-component ASNR values from 18.59 and 18.16 dB to 19.17 and 18.79 dB, respectively, and increased double-pulse ASNR from 20.62 to 21.10 dB. The subsequent amplitude-focused stage further reduced timing RMSE to 1.308 sampling periods and increased double-pulse

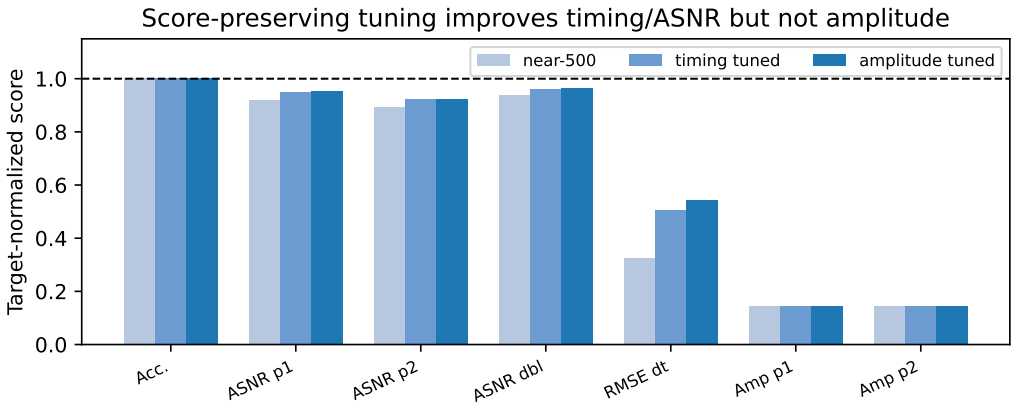


Figure 2. Target-normalized reconstruction performance of the continued checkpoints. Higher-is-better and lower-is-better metrics are normalized relative to the published reference values in opposite directions.

ASNR to 21.17 dB. Pulse-count accuracy changed only slightly across these stages, from 99.77% to 99.81%.

The behavior of amplitude RMSE differed from that of timing RMSE and ASNR. The second-component amplitude RMSE decreased from 0.3869 at the continued checkpoint to 0.3866 after timing-focused continuation and to 0.3859 after amplitude-focused continuation. The total reduction from the continued to the final checkpoint was therefore 0.0010, corresponding to approximately 0.26% of the continued-checkpoint value. Within the tested configuration range, the improvement in temporal localization and waveform agreement was not accompanied by a comparable improvement in amplitude estimation.

Figure 2 presents the same comparison after normalization by the published reference values. For higher-is-better metrics, the observed value is divided by the published reference value. For lower-is-better metrics, the published reference value is divided by the observed value. Values approaching or exceeding one therefore indicate closer agreement with, or improvement over, the corresponding published reference.

The final checkpoint exceeded the published pulse-count accuracy by 0.18 percentage points but did not reach the published reference for the reported waveform and component-level quantities. In particular, its timing RMSE remained above the published value of 0.710 sampling periods, and its second-component amplitude RMSE was approximately 6.9 times the published reference value of 0.0560. Because the local continuation and published reference originate from different training histories, these comparisons are descriptive and should not be interpreted as a paired statistical test. Nevertheless, the contrast between pulse-count accuracy and amplitude RMSE demonstrates that classification performance alone does not characterize component-level reconstruction fidelity.

5.2. Confidence calibration under controlled perturbations

The final continued checkpoint achieved an accuracy of 58.22% on the aggregate perturbed evaluation set. Temperature scaling did not change this value because it preserves the predicted class. Its effect was confined to confidence-dependent metrics, as summarized in Table 4.

Table 4. Aggregate classification and calibration results under controlled signal perturbations.

Probability output	Acc. (%)	ECE	NLL	$OER_{0.85}$ (%)
Uncalibrated	58.22	0.4011	8.5731	41.50
Temperature-scaled	58.22	0.3273	2.8681	39.69

Table 5. Calibration results for individual signal-perturbation conditions.

Perturbation	ECE_{uncal}	ECE_{temp}	NLL_{uncal}	NLL_{temp}
Noise only	0.4078	0.3288	8.168	1.993
Baseline only	0.3887	0.3026	7.000	1.591
Combined	0.3854	0.3016	7.812	2.268
Mild combined	0.4038	0.3227	7.529	1.566
Strong combined	0.4133	0.3428	9.134	3.468

Temperature scaling reduced ECE from 0.4011 to 0.3273 and NLL from 8.5731 to 2.8681. The overconfident error rate at the 0.85 threshold decreased from 41.50% to 39.69%, an absolute reduction of 1.81 percentage points. The unchanged accuracy confirms that these differences result from probability rescaling rather than from changes in pulse-count decisions.

Table 5 reports ECE and NLL separately for the five perturbation conditions. Temperature scaling reduced both metrics in every evaluated condition.

The strong combined condition produced the largest calibrated ECE and NLL among the reported conditions. However, the ordering was not monotonic across the remaining perturbation conditions. For example, the uncalibrated ECE for the reference combined condition was lower than that for the mild combined condition. The perturbation categories should therefore be interpreted individually rather than as a strictly ordered severity scale.

The baseline-only condition produced lower NLL than the noise-only condition before and after calibration. This observation describes the tested perturbation settings but does not establish that baseline variation is generally less disruptive than waveform noise, because the magnitudes of different perturbation types are not necessarily commensurate. Overall, temperature scaling consistently reduced confidence miscalibration under the evaluated conditions, but the remaining accuracy of 58.22% shows that it did not restore classification performance under perturbation.

5.3. Selective risk and coverage

Table 6 summarizes two operating points from the confidence-ranked evaluation. Retaining all perturbed traces resulted in an accuracy of 58.22% and a risk of 41.78%. Retaining the 50% of traces with the highest confidence increased selective accuracy to 71.91% and reduced selective risk to 28.09%.

At 50% coverage, selective accuracy was 13.69 percentage points higher than the accuracy obtained when all perturbed traces were retained. This result indicates that model confidence contains information about the relative likelihood of classification error within the evaluated perturbation set.

Table 6. Risk–coverage results for confidence-ranked predictions under controlled signal perturbations.

Selection rule	Coverage (%)	Selective accuracy (%)	Selective risk (%)
Retain all traces	100	58.22	41.78
Retain highest-confidence traces	50	71.91	28.09

For this binary classifier, scalar temperature scaling applies a monotonic transformation to the class-confidence margin and therefore does not alter the confidence ordering used for selection. The observed risk–coverage behavior should consequently be attributed to the ranking information in the original model confidence rather than to an improvement in ranking produced by temperature scaling. Calibration changes the numerical interpretation of confidence and the number of predictions exceeding a fixed confidence threshold, but it does not create the selective ordering reported in Table 6.

The 50% operating point is an empirical diagnostic rather than a prescribed acceptance threshold. It does not imply that half of the measured traces can be discarded without

affecting a spectroscopy analysis. It also does not provide a formal bound on future error rates or a conformal coverage guarantee. The result is restricted to the evaluated checkpoint, perturbation protocol, and confidence-ranking rule.

6. Discussion

The principal finding of this study is not a uniform improvement across all pulse-analysis quantities, but a separation among pulse-count classification, waveform reconstruction, component-parameter estimation, and confidence reliability. Under the nominal SAEFit protocol, continued training and configuration adjustment reduced timing RMSE from 2.177 to 1.308 sampling periods and increased double-pulse ASNR from 20.62 to 21.17 dB. Pulse-count accuracy remained high and reached 99.81%. In contrast, the reported second-component amplitude RMSE changed only from 0.3869 to 0.3859 and remained substantially above the published reference value of 0.0560. Under controlled signal perturbations, temperature scaling reduced ECE and NLL but did not change the shifted accuracy of 58.22%. These results demonstrate that improvements in one output of a multi-task pulse analyzer should not be generalized to its other outputs.

6.1. Reconstruction fidelity and component-parameter recovery

The parallel improvement in timing RMSE and ASNR indicates that the continued checkpoints produced closer temporal alignment and waveform agreement under the semi-synthetic evaluation protocol. However, the absence of a comparable reduction in amplitude RMSE shows that improved waveform agreement did not translate directly into accurate component-amplitude recovery. This distinction is important because timing, waveform reconstruction, and amplitude estimation are generated by related but non-equivalent objectives.

Several factors could contribute to this behavior. The amplitude-related loss may provide a weaker optimization signal than the timing and classification objectives, the parameter head may not adequately represent the ambiguity between overlapping components, or the semi-synthetic training distribution may insufficiently constrain amplitude combinations that produce similar summed waveforms. Correlation between component timing, amplitude, and pulse shape may also lead to multiple parameter combinations with similar reconstruction errors. The present experiments do not distinguish among these explanations. The results therefore indicate an amplitude-specific limitation within the tested architecture and configuration range, but they do not constitute a formal demonstration of non-identifiability.

The contrast between classification and amplitude estimation is particularly informative. Distinguishing a single pulse from an overlapped pulse is a global waveform decision, whereas recovering the amplitude of each constituent requires a more detailed decomposition of the observed signal. A model may consequently achieve near-saturated pulse-count accuracy while retaining substantial component-level error. The final checkpoint exceeded the published pulse-count accuracy but remained far from the published amplitude reference. Pulse-count accuracy should therefore not be used as a surrogate for timing or energy-related component fidelity.

The comparisons with the published 750-epoch values must nevertheless be interpreted cautiously. The published model, the local epoch-125 reproduction, and the continued checkpoints were obtained from different training histories. The published values provide a useful external reference, but the comparison is not equivalent to a controlled paired experiment. The present results support statements about improvement relative to the locally continued checkpoint; they do not establish general superiority over the original SAEFit model.

6.2. Confidence reliability under controlled perturbations

The reduction in ECE from 0.4011 to 0.3273 and in NLL from 8.5731 to 2.8681 shows that the uncalibrated softmax outputs were excessively sharp for the evaluated perturbation conditions. Temperature scaling reduced this sharpness while preserving the predicted class. The unchanged shifted accuracy is therefore expected and clarifies the nature of the improvement: calibration modifies the statistical interpretation of confidence but does not make incorrect classifications correct.

This distinction prevents calibration from being interpreted as robustness. An accuracy of 58.22% indicates that the selected classifier remained sensitive to the tested perturbations. Temperature scaling reduced confidence misreporting, including a 1.81-percentage-point reduction in the overconfident error rate, but it did not recover the nominal classification performance or improve the reconstructed pulse parameters. Similar distinctions between predictive accuracy and uncertainty reliability have been reported in broader evaluations of neural models under distribution shift [39,40].

Temperature scaling improved ECE and NLL under all five perturbation conditions, although the ordering of the conditions was not monotonic. This prevents the conditions from being interpreted as a simple severity sequence. In addition, the relative effects of noise and baseline variation depend on their chosen magnitudes and distributions. The lower NLL observed for the baseline-only condition therefore describes the present stress test and should not be generalized into a claim that baseline variation is inherently less disruptive than waveform noise.

The risk–coverage results provide a complementary observation. Retaining the highest-confidence 50% of perturbed traces increased selective accuracy from 58.22% to 71.91%. Thus, confidence ranking contained information about relative classification difficulty within the evaluated set. However, for binary classification, scalar temperature scaling applies a monotonic transformation to the class-confidence margin and does not change the ordering used for selection. The observed risk–coverage improvement should therefore be attributed to the ranking information already present in the model output rather than to a new ranking created by calibration.

The 50% coverage result is an empirical operating point, not a recommended universal threshold. Its practical suitability depends on the cost of deferring a trace, the availability of an alternative analysis route, and the effect of rejected or delayed events on the subsequent spectroscopy workflow. The present experiments do not quantify these downstream costs. A hybrid workflow in which lower-confidence traces are assigned to nonlinear fitting or expert review remains a plausible application, but its operational benefit requires direct evaluation rather than inference from selective accuracy alone.

6.3. Role of semi-synthetic supervision

The semi-synthetic construction provides an important but limited source of component-level supervision. Because each double pulse is formed by shifting, scaling, and summing measured single-pulse traces, the constituent timing and amplitude parameters are known by construction. The component waveforms also retain signal characteristics derived from detector measurements, which makes this protocol more representative than one based entirely on analytically generated pulse shapes.

Nevertheless, semi-synthetic overlaps are not equivalent to independently labeled measured double pulses. Their construction may not reproduce all dependencies among count rate, detector response, pulse shape, electronics behavior, baseline evolution, saturation, and sampling. It may also omit correlations that arise when naturally occurring events overlap under changing experimental conditions. Component-level RMSE obtained from this protocol therefore characterizes recovery under the specified construction rules.

The measured double-pulse traces provide evidence of naturally occurring waveform overlap but do not contain independently verified labels for each component. They cannot support supervised timing or amplitude RMSE without additional assumptions. For this reason, the present study does not transfer the observed semi-synthetic amplitude error directly to measured double pulses. Conversely, the absence of component labels should not be addressed through pseudo-labels derived from the model being evaluated, because doing so would make the reference dependent on the prediction.

The separation between semi-synthetic reconstruction evaluation and perturbed classification evaluation is therefore methodologically necessary. The former addresses component recovery when labels are available by construction, whereas the latter addresses confidence behavior under specified input changes. Neither experiment alone establishes accurate decomposition of naturally overlapped pulses under changing detector conditions.

6.4. Limitations and validity

Several limitations define the scope of the conclusions. First, the continuation study used a bounded configuration search involving existing learning rates and loss weights. The limited change in amplitude RMSE suggests that the tested controls were insufficient, but it does not exclude improvement through a broader search, an alternative loss formulation, or a different architecture.

Second, the reported continuation results are point estimates from the completed training trajectories. They do not provide multi-seed estimates of training variability. Small differences, such as the change in classification accuracy from 99.80% to 99.81%, should therefore not be interpreted as statistically meaningful. Repeated training would be required to distinguish systematic improvements from variation due to initialization, sampling, and checkpoint selection.

Third, model and checkpoint selection used feedback from the public evaluation protocol. The resulting checkpoint values should be interpreted as optimized outcomes within that protocol rather than as untouched estimates of performance on an independent detector population. Evaluation using an external detector run or a prospectively held-out dataset would provide a stronger test of generalization.

Fourth, the distribution-shift experiment was restricted to additive noise and baseline perturbations applied after normalization. Actual detector changes may also involve gain variation, pulse-shape changes, nonlinear electronics response, saturation, sampling artifacts, altered noise spectra, and changes in the distribution of temporal separation or pulse amplitude. Robustness to these untested conditions cannot be inferred from the current results.

Fifth, the calibration temperature was estimated using validation data drawn from the defined perturbation protocol. Its effectiveness under a qualitatively different or previously unseen shift remains unknown. Temperature scaling is also a global scalar correction and cannot represent input-dependent or class-dependent calibration errors.

Finally, the selective-prediction experiment provides an empirical risk–coverage relationship rather than a formal guarantee. It does not establish that a selected coverage will maintain the same error rate in a future detector run. The analysis also does not quantify the physical or operational consequences of deferring low-confidence events.

6.5. Future directions

The most immediate methodological priority is component-amplitude recovery. Future work should examine amplitude-specific objectives, parameterizations that explicitly preserve consistency between the reconstructed waveform and estimated component amplitudes, and constraints based on pulse area or other energy-related quantities. These

alternatives should be evaluated under the same component-labeled protocol and across repeated training runs to determine whether improvements are systematic.

A more decisive evaluation requires measured double pulses with independently established component timing and amplitude. Such labels might be obtained through controlled pulse injection, independently time-referenced event generation, or another experimental design in which the constituent signals are known before overlap. This would permit direct assessment of whether performance observed on semi-synthetic overlaps transfers to measured pile-up events.

The perturbation protocol should also be extended to detector-relevant changes in gain, pulse shape, baseline dynamics, noise spectrum, sampling, and count rate. Evaluation across independent acquisition runs or detector systems would help distinguish calibration that is specific to one stress construction from calibration that transfers across operating conditions.

More expressive uncertainty methods, including deep ensembles, may improve uncertainty representation but require repeated training and substantially greater computational cost [22]. Conformal methods could provide complementary coverage statements if their assumptions and calibration design are satisfied [41]. Neither approach should be assumed to correct weak amplitude decomposition; reconstruction quality and confidence reliability would still require separate evaluation.

Finally, a hybrid analysis path combining fast learned inference with nonlinear fitting for selected traces should be assessed at the workflow level. Such an assessment should report not only selective classification accuracy but also processing time, deferred-trace frequency, component-reconstruction error, and the effect on downstream spectroscopic quantities. This would determine whether confidence-based selection provides a measurable scientific benefit beyond improved calibration metrics.

7. Conclusions

This study examined whether reconstruction fidelity and confidence reliability improve concurrently in overlapped radiation-detector pulse analysis. Under the nominal SAEFit protocol, continued training reduced timing RMSE from 2.177 to 1.308 sampling periods and increased double-pulse ASNR from 20.62 to 21.17 dB, while pulse-count accuracy reached 99.81%. These improvements were not accompanied by comparable amplitude recovery: the reported second-component amplitude RMSE remained approximately 0.386, substantially above the published reference value of 0.056. Under controlled noise and baseline perturbations, temperature scaling reduced ECE from 0.4011 to 0.3273 and NLL from 8.5731 to 2.8681 without changing the shifted accuracy of 58.22%. At 50% coverage, the highest-confidence subset achieved a selective accuracy of 71.91%.

These findings support a clear distinction among pulse-count classification, component-level reconstruction, and confidence calibration. High classification accuracy does not establish accurate recovery of constituent amplitudes, and improved calibration does not correct classification or decomposition errors. The present evidence supports improved interpretation of model confidence under the tested perturbations, but it does not establish robustness to general detector-condition changes or accurate component recovery for measured double pulses. Further progress requires independently labeled measured overlaps, amplitude-specific modeling, evaluation under more representative detector variations, and repeated validation across training runs and acquisition conditions.

Author Contributions: writing—original draft preparation, T.H.; writing—original draft preparation, B.Z.; methodology, J.Z.; validation, Y.W; writing—review and editing, supervision, project administration, G.X. Z.G. and X.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Experimental data during this study are available from the corresponding author upon reasonable request.

Acknowledgments: The authors are grateful to all members of the research group for their valuable discussions and suggestions during this work.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Blank, B.; Borge, M. Nuclear structure at the proton drip line: Advances with nuclear decay studies. *Progress in Particle and Nuclear Physics* **2008**, *60*, 403–483. https://doi.org/10.1016/j.ppnp.2007.12.001.
2. Borge, M. Beta-delayed particle emission. *Physica Scripta* **2013**, *2013*, 014013. https://doi.org/10.1088/0031-8949/2013/T152/014013.
3. Hardy, J.C.; Towner, I.S. Superallowed 0+ to 0+ nuclear beta decays: 2014 critical survey. *Physical Review C* **2015**, *91*, 025501. https://doi.org/10.1103/PhysRevC.91.025501.
4. Larson, N.; Liddick, S.; Bennett, M.; et al. High efficiency beta-decay spectroscopy using a planar germanium double-sided strip detector. *Nuclear Instruments and Methods in Physics Research Section A* **2013**, *727*, 59–64. https://doi.org/10.1016/j.nima.2013.06.027.
5. Prokop, C.; Liddick, S.; Abromeit, B.; Chemey, A.; Larson, N.; Suchyta, S.; Tompkins, J. Digital data acquisition system implementation at the national superconducting cyclotron laboratory. *Nuclear Instruments and Methods in Physics Research Section A* **2014**, *741*, 163–168. https://doi.org/10.1016/j.nima.2013.12.044.
6. Prokop, C.; Liddick, S.; Larson, N.; Suchyta, S.; Tompkins, J. Optimization of the national superconducting cyclotron laboratory digital data acquisition system for use with fast scintillator detectors. *Nuclear Instruments and Methods in Physics Research Section A* **2015**, *792*, 81–88. https://doi.org/10.1016/j.nima.2015.04.052.
7. Chester, A.; Brown, B.A.; Burcher, S.P.; et al. Identification of a new isomeric state in 76Zn following the beta decay of 76Cu. *Physical Review C* **2021**, *104*, 054314. https://doi.org/10.1103/PhysRevC.104.054314.
8. Deng, Z.; Li, X.; Zhang, Y.; et al. Deep Learning-Based Signal Processing for Radiation Detection Systems. *Nuclear Instruments and Methods in Physics Research Section A* **2023**, *1055*, 168548. https://doi.org/10.1016/j.nima.2023.168548.
9. Sadeghi, B.; Scriven, D.; Chester, A.; Fox, R.; Liddick, S.; Cerizza, G. A machine learning framework for accurate and robust analysis of radiation detector pulses. *Nuclear Instruments and Methods in Physics Research Section A* **2026**, *1082*, 170971. https://doi.org/10.1016/j.nima.2025.170971.
10. Goodfellow, I.; Bengio, Y.; Courville, A. *Deep Learning*; MIT Press, 2016.
11. LeCun, Y.; Bengio, Y.; Hinton, G. Deep learning. *Nature* **2015**, *521*, 436–444. https://doi.org/10.1038/nature14539.
12. Gal, Y.; Ghahramani, Z. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. *International Conference on Machine Learning* **2016**, pp. 1050–1059.
13. Price, C.J.; Coope, I.D.; Byatt, D. A convergent variant of the Nelder–Mead algorithm. *Journal of optimization theory and applications* **2002**, *113*, 5–19.
14. He, K.; Zhang, X.; Ren, S.; Sun, J. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* **2016**, pp. 770–778. https://doi.org/10.1109/CVPR.2016.90.
15. Vaswani, A.; Shazeer, N.; Parmar, N.; et al. Attention is all you need. *Advances in Neural Information Processing Systems* **2017**, *30*.
16. Makansi, O.; Ilg, E.; Cicek, O.; Brox, T. Overcoming limitations of mixture density networks: A sampling and fitting framework for multimodal future prediction. In Proceedings of the Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2019, pp. 7144–7153.
17. Amini, A.; Schwarting, W.; Soleimany, A.; Rus, D. Deep Evidential Regression. In Proceedings of the Advances in Neural Information Processing Systems, 2020, Vol. 33, pp. 14927–14937.
18. Kingma, D.P.; Ba, J. Adam: A method for stochastic optimization. *International Conference on Learning Representations* **2015**.
19. Loshchilov, I.; Hutter, F. Decoupled weight decay regularization. *International Conference on Learning Representations* **2019**.
20. Paszke, A.; Gross, S.; Massa, F.; et al. PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems* **2019**, *32*.
21. Falcon, W.; team, T.P.L. PyTorch Lightning. *GitHub repository* **2019**.

22. Lakshminarayanan, B.; Pritzel, A.; Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in Neural Information Processing Systems* **2017**, *30*.
23. Nix, D.A.; Weigend, A.S. Estimating the mean and variance of the target probability distribution. *Proceedings of IEEE International Conference on Neural Networks* **1994**, pp. 55–60. https://doi.org/10.1109/ICNN.1994.374138.
24. Gawlikowski, J.; Tassi, C.R.N.; Ali, M.; Lee, J.; Humt, M.; Feng, J.; Kruspe, A.; Triebel, R.; Jung, P.; Roscher, R.; et al. A survey of uncertainty in deep neural networks: J. Gawlikowski et al. *Artificial intelligence review* **2023**, *56*, 1513–1589.
25. Platt, J.; et al. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in large margin classifiers* **1999**, *10*, 61–74.
26. Kuleshov, V.; Fenner, N.; Ermon, S. Accurate uncertainties for deep learning using calibrated regression. *International Conference on Machine Learning* **2018**, pp. 2796–2804.
27. Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K.Q. On calibration of modern neural networks. *International Conference on Machine Learning* **2017**, pp. 1321–1330.
28. Dawid, A.P. The well-calibrated Bayesian. *Journal of the American Statistical Association* **1982**, *77*, 605–610. https://doi.org/10.1080/01621459.1982.10477856.
29. Murphy, A.H. A new vector partition of the probability score. *Journal of Applied Meteorology* **1973**, *12*, 595–600. https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2.
30. Broecker, J.; Smith, L.A. Increasing the reliability of reliability diagrams. *Weather and Forecasting* **2007**, *22*, 651–661. https://doi.org/10.1175/WAF993.1.
31. Khaked, A.A.; Oishi, N.; Roggen, D.; Lago, P. In Shift and In Variance: Assessing the Robustness of HAR Deep Learning Models Against Variability. *Sensors* **2025**, *25*, 430. https://doi.org/10.3390/s25020430.
32. Goodarzi, P.; Schauer, J.; Schütze, A. Robust Distribution-Aware Ensemble Learning for Multi-Sensor Systems. *Sensors* **2025**, *25*, 831. https://doi.org/10.3390/s25030831.
33. Ghaleb, A.M.; Allahloh, A.S.; Mejjaouli, S.; Ali, M.A.; Al-Shayea, A. Uncertainty-Calibrated Safety Gating for Vision–Language–Action Manipulation Under Domain Shift: Reliability Gains and Intervention–Efficiency Trade-Offs. *Sensors* **2026**, *26*, 3140.
34. El-Yaniv, R.; et al. On the Foundations of Noise-free Selective Classification. *Journal of Machine Learning Research* **2010**, *11*.
35. Geifman, Y.; El-Yaniv, R. Selective classification for deep neural networks. *Advances in Neural Information Processing Systems* **2017**, *30*.
36. Vovk, V.; Gammerman, A.; Shafer, G. Algorithmic Learning in a Random World. *Springer* **2005**. https://doi.org/10.1007/b106715.
37. Shafer, G.; Vovk, V. A tutorial on conformal prediction. *Journal of Machine Learning Research* **2008**, *9*, 371–421.
38. Diao, R.; Zhou, M.; Meng, G.; Wang, S. Turbofan Engine Remaining Useful Life Prediction with Reliable Prediction Intervals via LSTM-Based Quantile Regression and Conformal Calibration. *Sensors* **2026**, *26*, 2249. https://doi.org/10.3390/s26072249.
39. Ovadia, Y.; Fertig, E.; Ren, J.; et al. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems* **2019**, *32*.
40. Hendrycks, D.; Dietterich, T. Benchmarking neural network robustness to common corruptions and perturbations. *International Conference on Learning Representations* **2019**.
41. Angelopoulos, A.N.; Bates, S. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *CoRR* **2021**, *abs/2107.07511*, [2107.07511].

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.